Shiv Kumar

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SUMMARY

Pre-final-year B.Tech (Data Science) candidate with hands-on experience in ML pipelines, full-stack development, and computer vision — focused on real-world deployment, clean data engineering, and measurable impact.

EDUCATION

Bachelor of Technology — Computer Science & Engineering (Data Science) | 2023 – 2027
Chaudhary Devi Lal University, Sirsa, Haryana | CGPA: 7.83 / 10 (Sem I–V)

TECHNICAL SKILLS

Languages	Python, R, C, C++
ML & AI	Scikit-learn, TensorFlow, Keras, XGBoost, OpenCV, MediaPipe
Modern AI & Tooling	LLMs & Prompt Engineering, LangChain, RAG, Agentic AI (watsonx), MLOps (Docker, CI/CD)
Web & Full Stack	Next.js, React, Tailwind CSS, FastAPI, Streamlit, jsPDF
Data & Visualization	Pandas, NumPy, Power BI, Tableau, Matplotlib, Seaborn
Cloud, DevOps & DB	IBM Cloud, Git, REST APIs, SQL, PostgreSQL, Supabase (RLS)

EXPERIENCE

Project Intern — Ransomware Simulator | May 2025 – Jun 2025
Centre for Development of Advanced Computing (CDAC), Noida, India

- Developed a controlled ransomware simulator in Python to model file-encryption attack patterns and lateral movement within isolated sandbox environments.
- Analyzed encryption-based attack vectors; documented recovery workflows and mitigation strategies for internal cybersecurity research.

AI Intern — Career-Mitar (Edunet Foundation × IBM) | Jun 2025 – Jul 2025
Edunet Foundation × IBM SkillsBuild — Remote

- Built Career-Mitar, an AI career-guidance agent (Flask + IBM watsonx.ai) performing resume analysis, skill-gap detection, roadmap generation, and interview prep.
- Engineered prompt pipelines against watsonx foundation models; shipped a working prototype demonstrating end-to-end career coaching flows.

PROJECTS

AeroSpeed Analytics — F1 Telemetry & Road Car Aerodynamics | 2025
Personal R&D project

Tech: Python, FastF1, XGBoost, Streamlit, FastAPI

Links: github.com/Shivmahlan/aerospeed-analytics · kaggle.com/shivmahlan · Live Demo

- Engineered a dual-domain dataset of 60K+ F1 telemetry laps (2022–2024) fused with EPA fuel-economy and scraped drag-coefficient data; published on Kaggle.
- Trained three ML models (Random Forest, XGBoost, Gradient Boosting) with cross-validated pipelines; deployed a 7-page Streamlit dashboard with FastAPI backend to Streamlit Cloud.

Real-Time Hand Gesture Control System | 2024 – Present
Personal project

Tech: Python, OpenCV, MediaPipe, PyAutoGUI

Links: github.com/Shivmahlan/hand-gesture-control

- Built a dual-hand pipeline at 720p/30+ FPS: right hand for cursor navigation, left for scroll — fully touch-free desktop interaction.
- Implemented a virtual 60% QWERTY keyboard via a custom rule-based gesture engine on coordinate geometry and landmark thresholds — no pre-trained classifiers.

Dairy Management SaaS Platform | 2024 – Present
Deployed across multiple organizations

Tech: Next.js, React 19, Tailwind CSS, Supabase, PostgreSQL, jsPDF

Links: github.com/Shivmahlan/dairy-manager

- Engineered a multi-tenant web app for dairy businesses to manage milk collection, financial ledgers, and inventory — deployed across multiple organizations.
- Architected a business-scoped PostgreSQL schema on Supabase with Row-Level Security (RLS) for strict data isolation; built real-time milk collection with dynamic pricing by weight, fat content, and business-specific rates.
- Designed an automated 10-day financial ledger with carry-forward balances and closed-cycle protection to prevent tampering in settled periods.
- Delivered measurable impact: cut manual recordkeeping by ~1 hour per 6-hour shift via automated entry, PDF reports (jsPDF), and bulk CSV import/export.

dcleaner — Python Data-Cleaning Library | 2025

Published on PyPI | Author & maintainer

Tech: Python, pandas, CLI

Links: github.com/Shivmahlan/dcleaner · pypi.org/project/dcleaner

- Authored and published dcleaner, a fluent, low-boilerplate data-cleaning and visualization library on top of pandas.
- Designed a chainable API for common cleaning workflows (dedupe, type coercion, missing-value handling); distributed via PyPI with CI and documentation.

Career-Mitar — AI Career-Guidance Agent | 2025

Built during Edunet × IBM internship

Tech: Python, Flask, IBM watsonx.ai

Links: github.com/Shivmahlan/Career-mitar

- Built an AI agent that analyzes a resume, detects skill gaps, and generates personalized learning roadmaps and interview-prep plans.
- Integrated IBM watsonx.ai foundation models via prompt pipelines; delivered a working prototype with resume upload and structured coaching output.

LEADERSHIP & ACTIVITIES

Technical Coordinator — InfoMania X | 2024 – Present

Technical Club, Chaudhary Devi Lal University, Sirsa, Haryana

- Coordinated and conducted multiple technical events; managed end-to-end logistics, participant outreach, and cross-team execution.

CERTIFICATIONS

- Getting Started with Artificial Intelligence — IBM SkillsBuild, Jun 2026
- Getting Started with Cybersecurity — IBM SkillsBuild, Jun 2026
- Data Science & Machine Learning — Internshala, Jul 2025
- Oracle Database & Cloud Foundations — Oracle, Sep 2025
- Career Essentials in Data Analytics — Microsoft / LinkedIn Learning, Feb 2025